AGILE BACKLOG CREATION & SPRINT SIMULATION

Jira Lab 2 — Submission Report

PUBLIC BUS LIVE TRACKING & CROWDING ESTIMATOR

Project	Public Bus Tracking System
Agile activity	Backlog creation, estimation and sprint simulation
Sprints completed	Sprint 1 and Sprint 2
Total estimated work	61 story points

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Prepared from the Jira screenshots and Lab 1 / Lab 2 materials provided.

1. Lab Overview

This report documents the Jira implementation for the Public Bus Tracking System. The work converts the functional requirements from Lab 1 into Agile Epics and User Stories, assigns relative story-point estimates, plans and simulates two one-week sprints, and reviews progress using Jira's sprint and burndown reports.

The Jira backlog was organized around functional themes visible in the project: Live Bus Tracking & ETA, Commuter Bus Information, Crowding Estimation, Fleet Monitoring & Alerts, and System Performance & Security.

Submission Evidence

Required evidence	Included	Evidence in this report
Jira backlog with Epics and User Stories	**Yes**	Section 2
Story point assignments	**Yes**	Section 3
Sprint board — Active Sprint view	**Yes**	Section 4
Burndown Chart	**Yes**	Section 5
Brief reflection answers	**Yes**	Section 6

Project-to-Jira Mapping

The Lab 1 requirements identify actors such as the Commuter, Fleet Controller, Bus GPS Unit, and Passenger Counting Sensor, with use cases including viewing live bus status, checking stop ETA, viewing/calculating crowding, monitoring the fleet, receiving GPS data, receiving passenger count, sending service alerts, and obtaining the latest bus location. These themes are represented in the Jira backlog.

2. Jira Backlog — Epics, User Stories & Organization

The following Jira backlog screenshot shows the project stories, their Epic labels, workflow status, and story-point estimates. The Epic labels group related stories by functional theme.

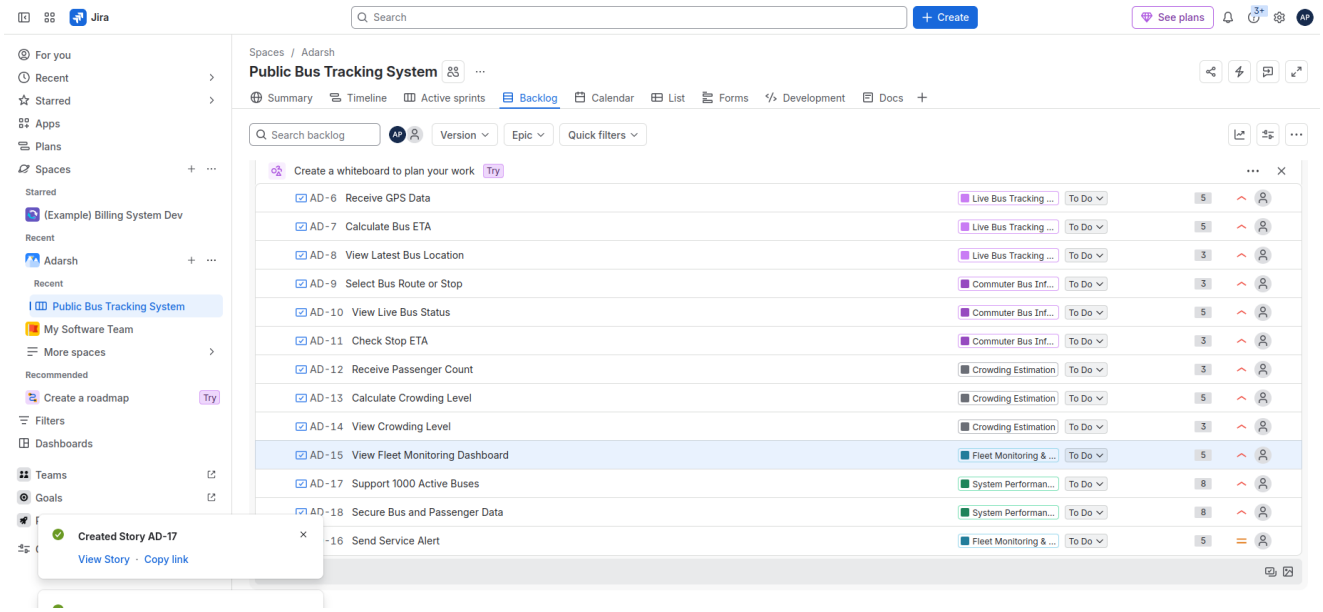


Figure 1. Jira backlog showing User Stories grouped by Epic labels and their story points.

A second view shows an individual story together with its Parent Epic, demonstrating the Epic-to-Story relationship used in the backlog.

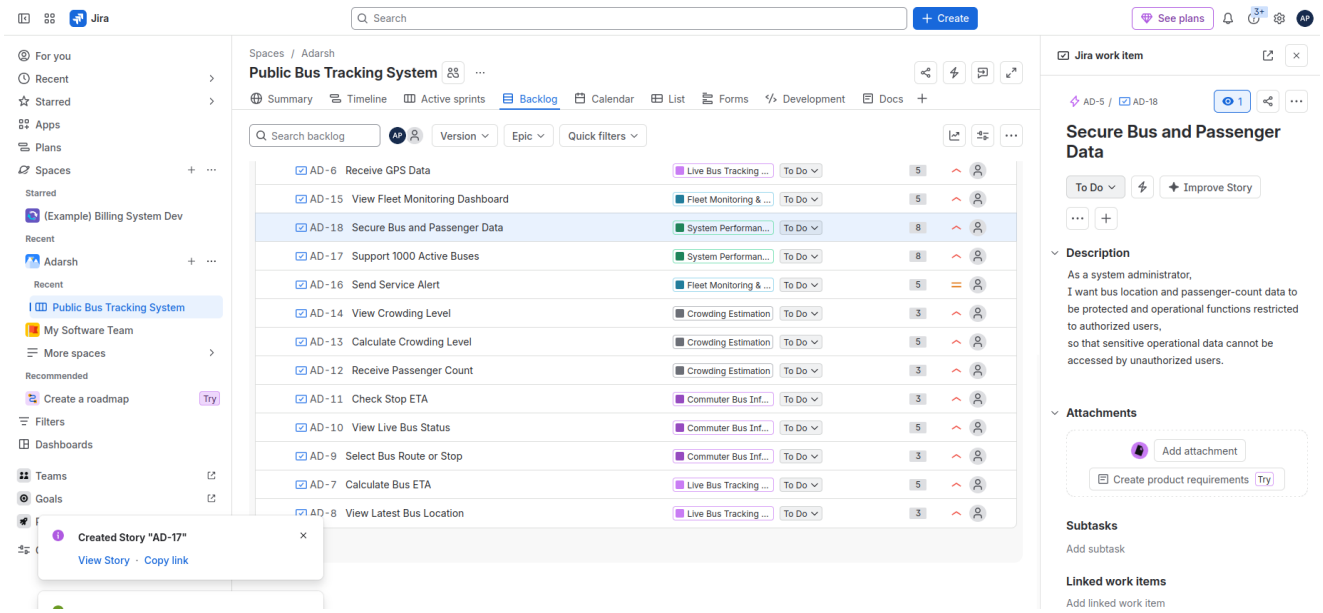


Figure 2. Jira work-item view showing 'Secure Bus and Passenger Data' under its parent Epic.

Observed Epic Themes

Epic / Theme	Representative purpose
Live Bus Tracking & ETA	GPS reception, bus ETA calculation and latest bus-

	location functionality
Commuter Bus Information	Route/stop selection, live bus status and stop ETA
Crowding Estimation	Passenger-count reception, crowding calculation and crowding display
Fleet Monitoring & Alerts	Fleet dashboard and service alerts
System Performance & Security	Scalability and protection of bus/passenger data

3. Story Point Assignments

Story points were assigned using the Fibonacci-style scale described in the lab handout. The estimates represent relative effort, complexity, and uncertainty rather than hours.

Issue	User Story	Story Points
AD-6	Receive GPS Data	5
AD-7	Calculate Bus ETA	5
AD-8	View Latest Bus Location	3
AD-9	Select Bus Route or Stop	3
AD-10	View Live Bus Status	5
AD-11	Check Stop ETA	3
AD-12	Receive Passenger Count	3
AD-13	Calculate Crowding Level	5
AD-14	View Crowding Level	3
AD-15	View Fleet Monitoring Dashboard	5
AD-16	Send Service Alert	5
AD-17	Support 1000 Active Buses	8
AD-18	Secure Bus and Passenger Data	8
	Total estimated backlog	61

Jira Evidence

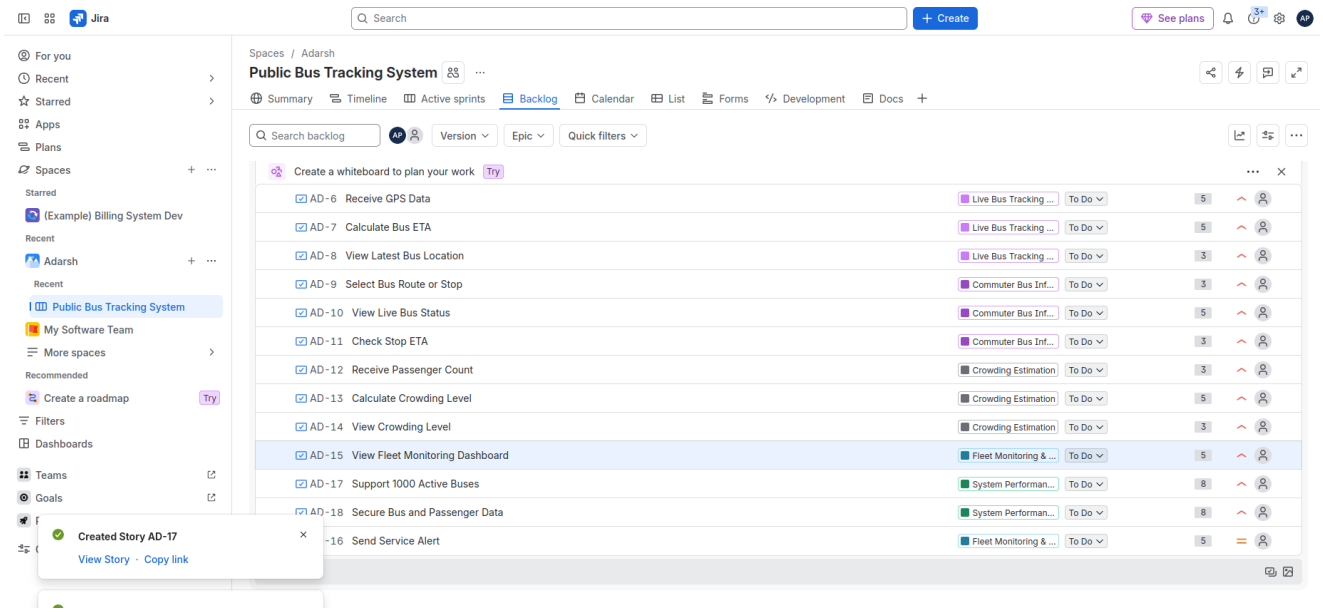


Figure 3. Story-point values visible in the Jira backlog.

4. Sprint Planning & Active Sprint Simulation

The lab required two one-week sprints and a simulated progression of work through To Do → In Progress → Done. The Jira screenshots show Sprint 1 and Sprint 2 being planned and the Active Sprint board being used to move stories.

Sprint 1: 6 stories, 24 story points.

Sprint 2: 7 stories, 37 story points.

Total across the two sprints: 61 story points.

Figure 4a. Sprint 1 planned work in the Jira backlog.

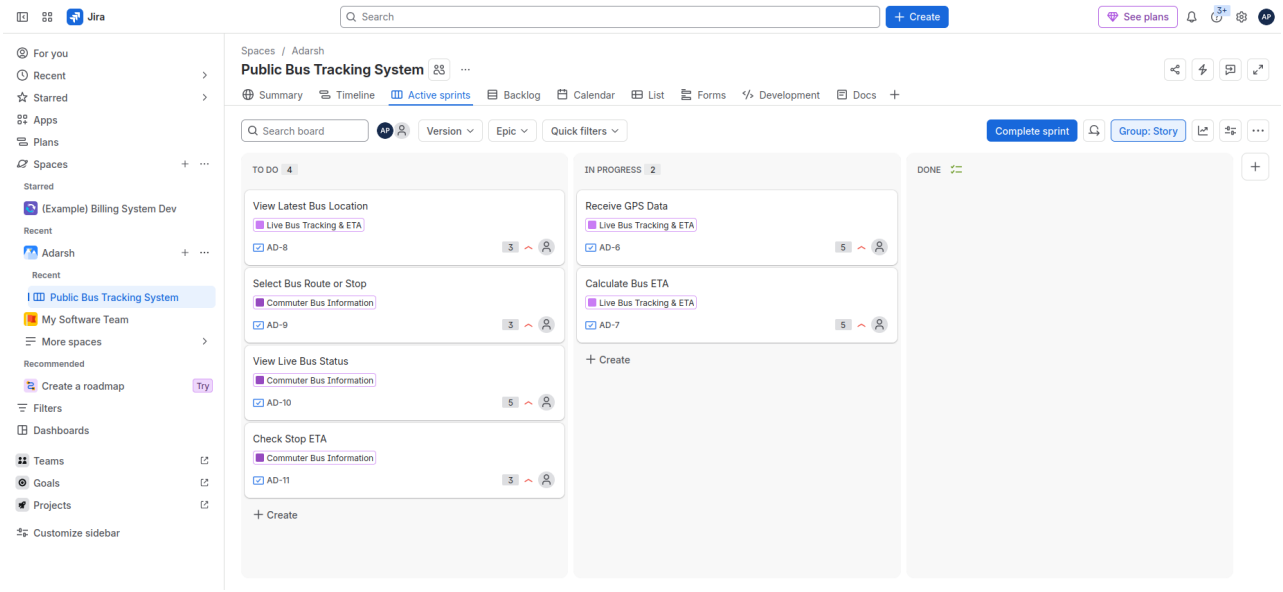
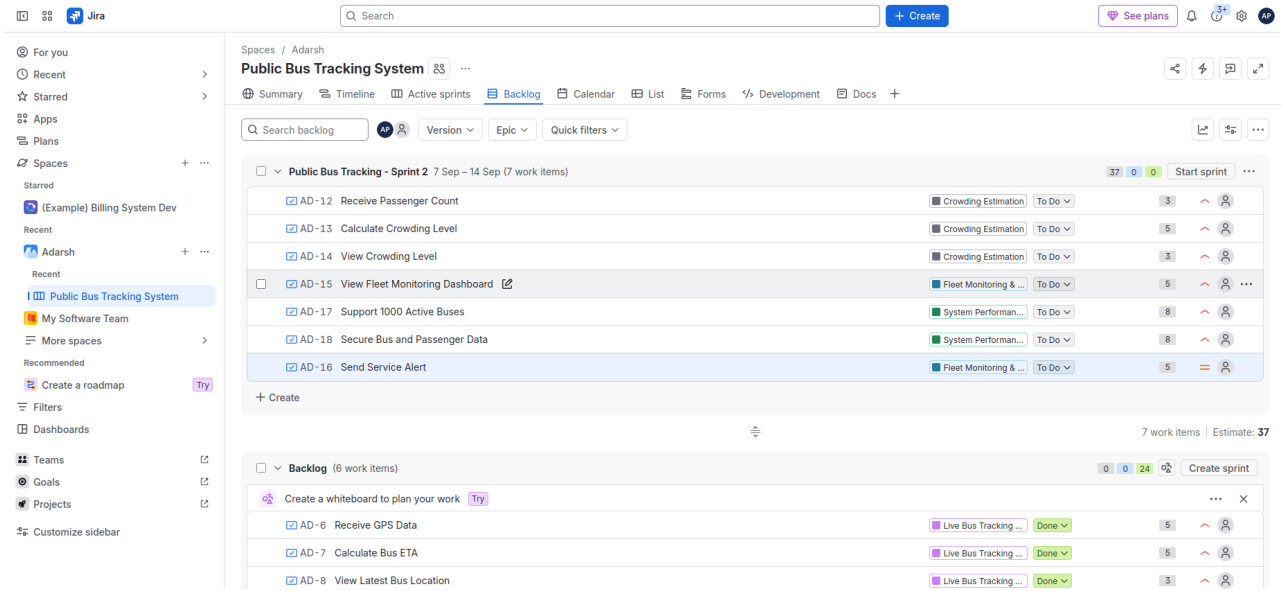


Figure 4b. Sprint 2 planned work in the Jira backlog.



These two backlog views show the planned distribution of work between Sprint 1 (24 story points) and Sprint 2 (37 story points).

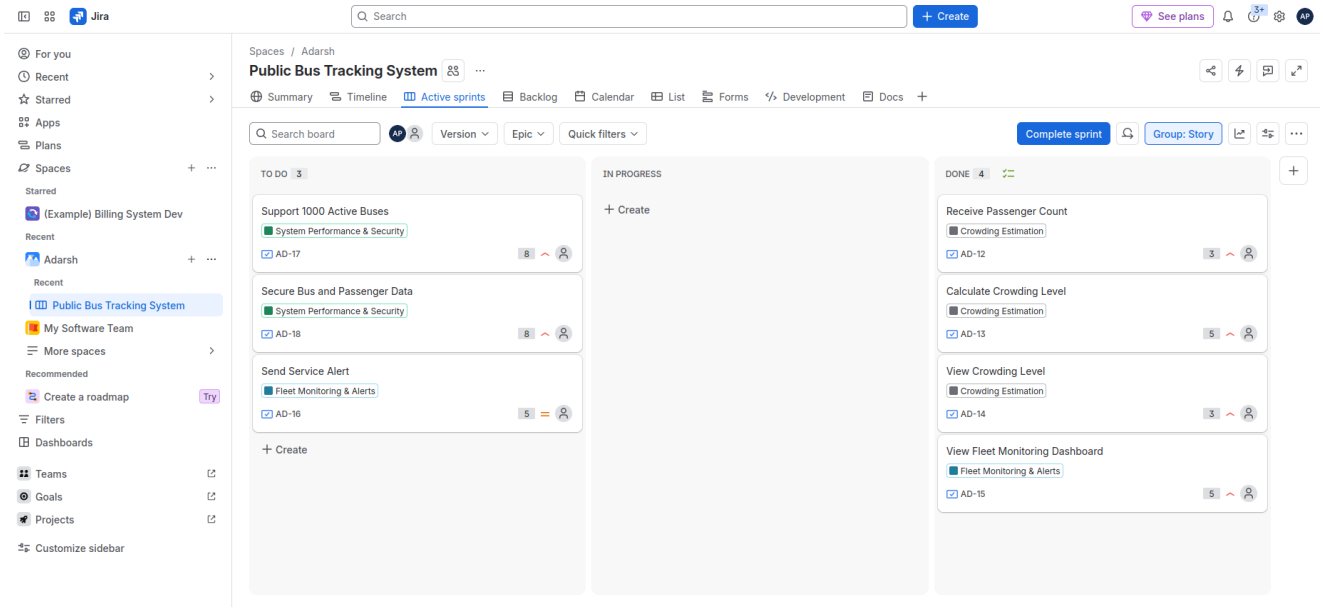


Figure 4c. Jira Active Sprint board showing stories distributed across To Do, In Progress and Done.

The board demonstrates the simulated workflow: some stories are in progress while others have already been completed. The exercise therefore provides evidence of sprint execution rather than a real production development cycle.

5. Burndown Chart

Jira's Burndown Chart was generated for Sprint 2 using Story Points as the estimation statistic. The chart starts from 37 story points, matching the Sprint 2 estimate shown in the backlog.

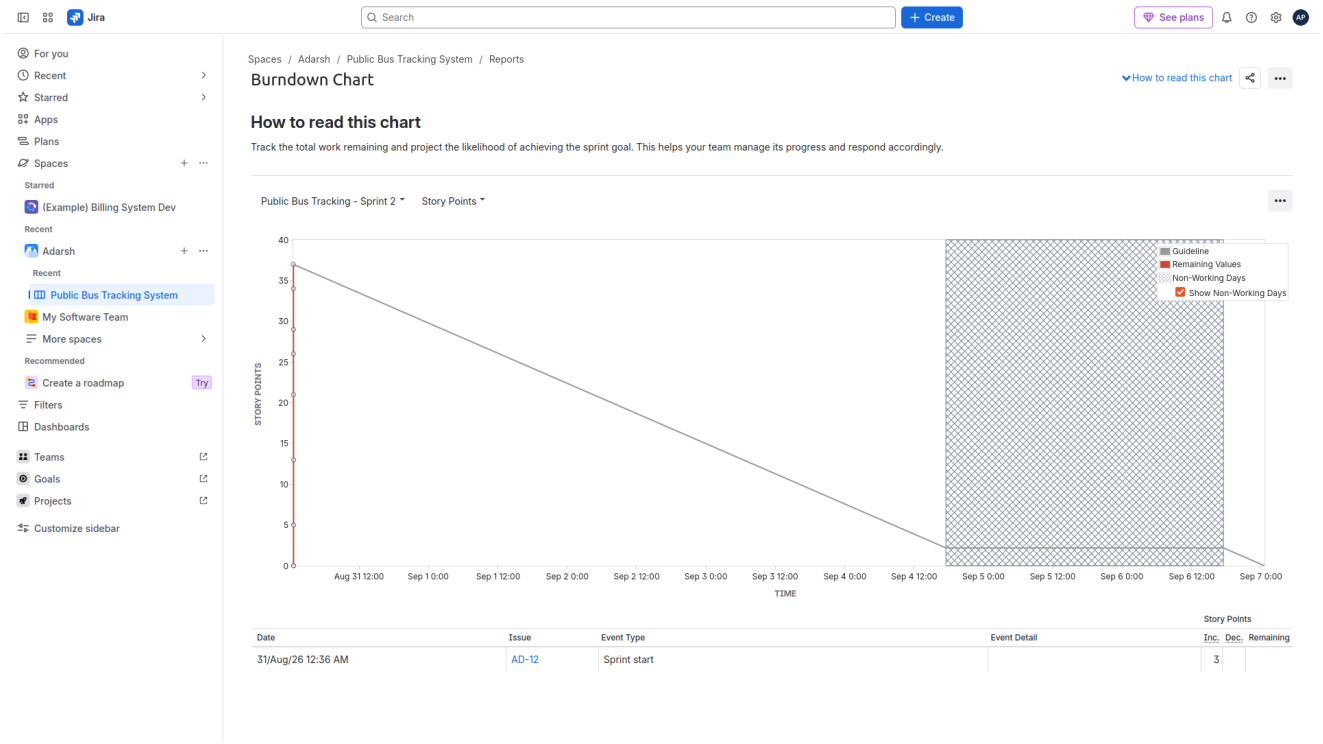


Figure 5. Jira Burndown Chart for Public Bus Tracking — Sprint 2.

The chart contains the ideal Guideline and the Remaining Values series. Because the exercise was a simulation performed over a very short period, the chart should be interpreted as evidence of how Jira visualizes sprint capacity and progress rather than as a statistically meaningful velocity measure.

Sprint Report Evidence

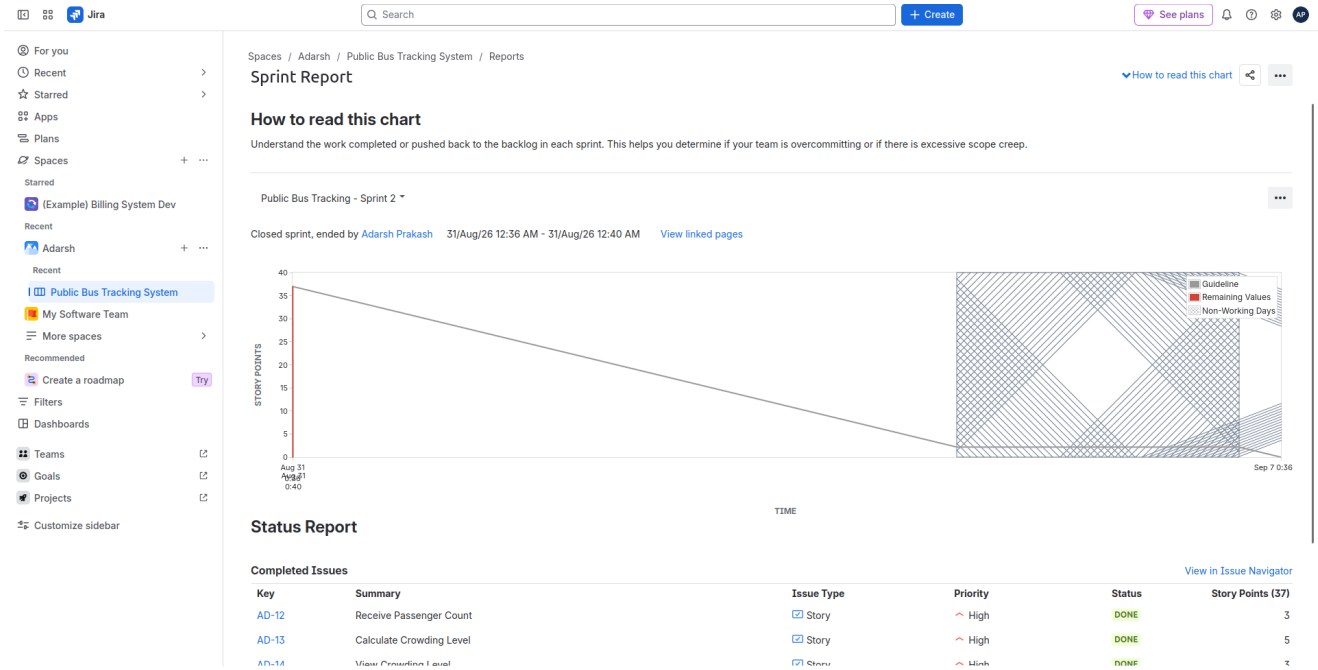


Figure 6. Jira Sprint Report showing Sprint 2 completion and the 37-story-point estimate.

6. Reflection

The following answers are based on the Jira simulation and the Lab 2 instructions. Since this was a simulated sprint, the reflection distinguishes estimated effort from real measured development effort.

1. Did your estimations reflect the actual effort?

The estimates were reasonable as relative effort estimates, but they cannot be fully validated against actual effort because the lab was a simulated sprint and no real development hours or team velocity were recorded. The Fibonacci values still provided a useful distinction between smaller stories, such as View Latest Bus Location (3 points), and more complex work, such as Support 1000 Active Buses and Secure Bus and Passenger Data (8 points).

2. Was your backlog well-prioritized?

Overall, the backlog was organized around the core functionality of the public bus tracking system and divided into two sprints. The earlier sprint focused on fundamental tracking and commuter-information capabilities, while the later sprint covered crowding estimation, fleet monitoring, alerts, scalability and security. This provided a sensible progression from core user-facing tracking functionality toward supporting and non-functional concerns.

3. How did your simulated sprint align with your plan?

The simulation followed the planned Scrum workflow. Stories were selected into one-week sprints and then moved through To Do, In Progress and Done on the Active Sprint board. Sprint 1 contained 24 story points and Sprint 2 contained 37 story points. The Jira board and reports therefore show that the planned backlog was translated into sprint work and tracked to completion.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart showed the amount of estimated work remaining against the ideal guideline. Sprint 2 began with 37 story points, so the chart provided a visual reference for how quickly the team would need to burn down work to meet the sprint goal. The main insight from this exercise is that burndown charts help compare planned progress with actual remaining work and can highlight whether a team is on track, overcommitted, or completing work at a different rate than expected. Because this was a simulation, it should not be treated as a reliable measurement of long-term team capacity or velocity.

Conclusion

The Jira exercise demonstrated the complete basic Agile workflow: converting requirements into Epics and User Stories, estimating and prioritizing backlog items, planning two sprints, simulating progress on an Active Sprint board, and using Jira reporting to review progress. The resulting evidence covers all items listed in the submission guidelines.